VIBRATION—SHAKE—RESONANCE NOISE—SPEEDS ABOVE 72 KM/H (45 MPH)	Article No. 00-14-3

FORD: 1999-2001 EXPLORER

ISSUE

A vibration at speeds above 72 km/h (45 mph) may occur on some vehicles. This may be due to an imbalance in one or more of the driveline components which results in a resonance that produces a noticeable vibration usually in the seat or steering column.

ACTION

Refer to the following Drive Diagnostic Procedure and System Balancing Procedures. Follow the initial drive diagnostic and note any changes after each drive.

DRIVE DIAGNOSTIC PROCEDURE

1. Ask questions of the customer and find out when the vibration occurs. Base diagnostic work on customer usage of vehicle (such as gear, speed, load, and towing.) It is also best to ride with the customer and make sure you know exactly what the concern is. Also, use an Electronic Vibration Analyzer (EVA) if available at seat track, rear pinion, and front differential cover to find the source of the concern.
2. Start the vehicle. Does a vibration exist when you perform neutral rpm engine run-ups? If it does, review engine driven components, fan, and transmission torque converter for possible imbalances. Repair these concerns first and retest.
3. Before driving the vehicle, review the tires for signs of wheel misalignment or worn suspension parts such as shock absorbers, ball joints, and bushings. (Reference Section 100-04 in the appropriate model year Workshop Manual.) Repair and/or replace worn parts. Drive vehicle to see if concern is corrected.
4. Document the operating conditions required to duplicate the customer's concern including: Coast/Cruise, AUTO/4H/4L, Acceleration/Deceleration.
5. Document the vehicle speed, engine rpm, and gear at which the concern is duplicated. If an EVA is available, document the frequency (hertz) and amplitude of the vibration.
6. On 4.0L 4X4 models while at a stop, place the transmission in Neutral. Switch from 4AUTO to 4L. Place the vehicle in Drive and drive a short distance (6-9 meters/20-30 feet). Put the transmission back in Neutral and switch back to 4AUTO. Re-drive the vehicle at highway speed. If the vibration goes away or gets worse, contact Borg-Warner at 1-765-286-6569.
7. Inspect transmission mount to see if it is twisted or mispositioned. Look for strap grounding out against the mount. If this is occurring, loosen the mount bolts, neutralize the mount position, and allow the strap to straighten out. Tighten the nuts and bolts to 87-110 N•m (64-81 lb-ft) in torque stages. Retest the vehicle.
8. If an EVA is available, isolate the vibration to a probable rotating component. Using the EVA, match the monitored frequency to a possible rotating component using the engine speed or vehicle speed method as described in Section 3-30 of the NVH Student Reference Book supplied with the Ford NVH class or Page 0-9 of the SPX Kent-Moore Instruction Manual that comes with the EVA test equipment.
9. After recording all information, review the following Symptom Chart first to avoid the replacement of good parts.

SYMPTOM CHART

CONCERN	POSSIBLE SOURCE	ACTION
Shimmy	Tire Steering Axle Flange Alignment Rear Suspension Sag	Go to Pinpoint Test A1
Body Boom	Driveline Driveshafts Body Resonance Exhaust System	Go to Pinpoint Test B1
Vehicle Speed Dependent Vibration	Tire/Wheel Driveshaft Axle Transfer Case	Go to Pinpoint Test B1
Engine Speed Dependent	Engine Drive Accessories FESM Mounts Body Mounts Engine Mounts Exhaust System	Go to Pinpoint Step C1

SYSTEM BALANCING USING AN EVA

The following procedure is a basic outline on how to system balance the vehicle using various methods available. For more detailed information, refer to Section 100-04 of the Workshop Manual or the manual provided with your Electronic Vibration Analyzer (EVA).

Before starting, be sure that the runout of the driveline components is within specification. This specification can be found in the Workshop Manual or in this Diagnostic Procedure. If components are found to be good, proceed with strobe balancing using the EVA.

1. Number the ends of the driveshaft by the transmission and the axle pinion 1 through 4, putting a number every 90 degrees. Then place an accelerometer at each end of the driveshaft.
2. Mount the accelerometer on the bottom of the rear axle differential or the transmission extension housing. Be sure to position the sensor as close to the driveshaft as possible, assuring that the UP side of the accelerometer is up and the sensor is horizontal.
3. Have the vehicle positioned on a hoist and set up on jack stands to level the vehicle with the wheels removed.

NOTE

BEFORE STARTING TO STROBE THE VEHICLE, BE SURE ALL LOOSE ARTICLES ARE CLEAR OF MOVING COMPONENTS. NEVER TOUCH MOVING PARTS WITH YOUR HANDS.

4. Turn off all accessories and have an associate start the engine. Place the engine in gear and accelerate the vehicle to speed where the vibration occurs.
5. With an inductive timing light clipped to the trigger wire of the EVA and the accelerometer plugged into input A, be sure both the EVA and timing light have a 12 volt source.
6. Verify that the predominant frequency displayed on the EVA matches the frequency of the original reading taken with the EVA. Press "STROBE" key on the EVA pad. The EVA will display a series of questions in order to select the correct filter. Press "YES" to select the correct filter. Be sure that the frequency is in the middle of the filter range. Use full range only if these ranges do not give the desired results.

7. The display on the EVA shows the test frequency, the amplitude, and the filter range. The driveline is balanced when the strobe amp is between 0-4, or when the flashing strobe becomes erratic or ceases.

8. Point the timing light at the driveshaft. The strobe will appear to freeze the driveshaft. Note at which of the numbered marks it is on or closest to.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 303000, 304000, 702000, 702300, 703000, 703200, 703300

Test Step		Result ▶	Action to Take
A1	SET TIRE PRESSURE		
	• Set Explorer tires to recommended pressure shown on tag located on the driver's door. • Redrive vehicle. • **Does vehicle still exhibit concern?**	Yes ▶ No ▶	GO to **A2**. Return vehicle to customer.
A2	INSPECT WHEELS AND TIRES		
	• Check tire for unusual wear or damage. If these conditions exist, replace tire. **NOTE: REFER TO TSB 98-5-2 FOR TIRE COMPANY CONTACT PHONE NUMBERS.** • Check wheels for missing balance weights. If any wheels are missing weights, dynamically balance wheels and tires and recheck air pressure. • Reinstall and torque lug nuts to 150 N•m (110 lb-ft). • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **A3**. Return vehicle to customer.
A3	TIRE ROAD FORCE MEASUREMENT		
	• **Does your dealership own, or have access to a tire road force measurement machine?** • The most common example is the Hunter GSP9700.	Yes ▶ No ▶	GO to **A11**. GO to **A4**.
A4	SWAP TIRES WITH KNOWN GOOD VEHICLE		
	• If another Explorer or Ranger that does not exhibit a vibration is available, place all four (4) tire/wheel assemblies from the good vehicle onto current vehicle. Be sure to set tire pressures to the specification listed on driver's door label. • Mark each of the original wheels with the location on the vehicle where it was mounted (i.e., LF, LR, RF, RR). • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B1**. GO to **A5**.
A5	MEASURE RUNOUT OF HUB ASSEMBLY		
	• Reference Pinpoint Tests J7 and J10 in Section 100-04 of the Workshop Manual. • Measure the hub pilot or bolt circle radial runout. • **Is pilot radial runout less than 0.152mm (0.006″)?** • **Is bolt radial circle runout less than 0.381mm (0.015″)?**	Yes ▶ No ▶	GO to **A6**. REPAIR or install new components as necessary.

TB-6148-A

Figure 1 - Article 00-14-3

Test Step		Result ▶	Action to Take
A6	**ROTATE** ORIGINAL FRONT TIRES		
	• Remove the rear tires currently installed on the vehicle. Install the original front tires back onto the truck at the rear of the vehicle. • Redrive vehicle. • **Did the vibration return?**	Yes ▶ No ▶	GO to **A8**. GO to **A7**.
A7	REINSTALL ORIGINAL REAR TIRES ON FRONT		
	• Remove the front tires currently installed on the vehicle. Replace them with the original rear tires. • Redrive vehicle. • **Did the vibration return?**	Yes ▶ No ▶	GO to **A8**. Return vehicle to customer.
A8	SWAP THE TIRES		
	• Swap the tires one at a time with the known "good" tires to isolate the vibration to one or more tire/wheel assemblies. • **Is the concern isolated to one or more tire/wheel assemblies.**	Yes ▶ No ▶	GO to **A9**. GO to **B1**.
A9	MEASURE RADIAL RUNOUT OF TIRE AND WHEEL		
	• Use procedure outlined in the Workshop Manual, Section 204-04. Allowable tire/wheel assembly radial runout is 1.143mm (0.045″). Allowable wheel radial runout is 0.635mm (0.025″). • **Is tire and wheel within specification?**	Yes ▶ No ▶	GO to **A10**. REPLACE tire and wheel. Follow Workshop Manual, Section 204-04. Redrive to verify repair. Return vehicle to customer.
A10	REMOUNT TIRE TO WHEEL		
	• Using a grease pencil, mark the current relationship of the tire to the wheel. • Deflate the tire and rotate the tire 90 degrees relative to the wheel. • Dynamically balance the tire and recheck air pressure. • Reinstall and torque lug nuts to 150 N•m (110 lb-ft). • **Does concern still exist?**	Yes ▶ No ▶	REPEAT Step A10 at 180 and 270 degrees. If this does not fix the concern, replace tire and wheel. Redrive to verify repair. Return vehicle to customer. Return vehicle to customer.

TB-6149-A

Figure 2 - Article 00-14-3

Test Step		Result ▶	Action to Take
A11	MEASURE TIRE ROAD FORCE VARIATION		
	• Using procedure from Instruction Manual for tire road force measurement machine, measure road force variation of all four (4) tires. • **Does any tire/wheel assembly measure more than 6.8 kg (15 lbs) for R1H, R2H, or R3H?** **NOTE: HUNTER DEFAULT SPECIFICATION IS 12 kg (26 lbs). USE 6.8 kg (15 lbs) SPECIFICATION FOR EXPLORER/MOUNTAINEER VEHICLE LINE.**	Yes ▶ No ▶	GO to **A12** . GO to **B1** .
A12	REMOUNT TIRE TO WHEEL		
	• Using tire road force measurement machine, determine low spot of tire radial runout. • Match high point for force variation with low spot of tire radial runout. • Retest tire. • **Is tire now within specification?**	Yes ▶ No ▶	REMOUNT tires and redrive vehicle. If concern is eliminated, return vehicle to customer. REPLACE tire. Follow Workshop Manual, Section 204-04. Redrive to verify repair. Return vehicle to customer.

TB-6150-A

Figure 3 - Article 00-14-3

Test Step		Result ▶	Action to Take
B1	VISUALLY INSPECT DRIVELINE		
	• **Does driveshaft have any dents or damage?** If so, replace driveshaft with proper service part. GO to **B6**. • Inspect driveshafts, front and rear axles, and transmission for loose components. • Tighten loose components. • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B2**. Return vehicle to customer.
B2	CHECK ALIGNMENT OF DRIVESHAFT		
	• Both driveshaft and axle flange(s) will have yellow dots painted on them. Verify that the driveshaft and axle flange dots are aligned. • **Are yellow dots aligned?** **NOTE: 2001 MODEL YEAR EXPLORER SPORT AND EXPLORER SPORT TRAC (4X4) ALSO REQUIRE DRIVESHAFT ALIGNMENT TO TRANSFER CASE.**	Yes ▶ No ▶	GO to **B4**. GO to **B3**.
B3	REALIGN DRIVESHAFT		
	• **Does the condition still exist when the driveshafts are realigned?**	Yes ▶ No ▶	GO to **B4**. Return vehicle to customer.
B4	MEASURE DRIVESHAFT RUNOUT		
	• Measure driveshaft runout at both edns and in the middle using a dial indicator. The runout must not exceed 0.762mm (0.030″). • **Does runout exceed 0.762mm (0.030″) at any location?**	Yes ▶ No ▶	GO to **B5**. GO to **B7**.
B5	RE-INDEX DRIVESHAFT		
	• Reinstall driveshaft in the vehicle, matching the high point of the measured runout with the yellow dot on the axle. • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	REPLACE driveshaft with proper service part. GO to **B6**. Return vehicle to customer.
B6	REDRIVE WITH NEW DRIVESHAFT		
	• Redrive vehicle with new driveshaft after ensuring runout is within specification and all driveline components are properly aligned to painted dots. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B7**. Return vehicle to customer.

TB-6151-A

Figure 4 - Article 00-14-3

Test Step		Result ▶	Action to Take
B7	MEASURE RADIAL RUNOUT AT REAR AXLE FLANGE		
	• Measure radial runout of rear axle flange using a dial indicator and companion flange runout gauge 205-319 and clamp plate 205-320 (Rotunda Tools) to check for accuracy of yellow dot. If not accurate, re-mark at high point and align driveshaft to the proper high point. If runout is over 0.203mm (0.008″), replace axle. • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B8** . Return vehicle to customer.
B8	VEHICLE DRIVE CONFIGURATION		
	• **Is the vehicle a 4X2 or 4X4 or AWD?**	4X2 ▶ 4X4/AWD ▶	GO to **B9** . GO to **B12** .
B9	INDEXING OF DRIVESHAFT AXLE SIDE		
	• Before disconnecting driveshaft, mark one of the shaft mounting bolt holes "A" and number the pinion flange 1 through 8. Starting with 1A, rotate the driveshaft 180 degrees to A5 and road test. • Does this improve or eliminate vibration? If not, re-index to other seven (7) positions. Find the position where the vibration is gone or most improved. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B10** . Return vehicle to customer.
B10	FIELD BALANCING OF DRIVESHAFT		
	• Field balance driveshaft per Workshop Manual, Section 205-00. (Even with a new driveshaft, you should do the field balancing as outlined.) **NOTE: TO PERFORM SYSTEM BALANCE USING AN ELECTRONIC VIBRATION ANALYZER (ROTUNDA 014-00344), FOLLOW TECHNIQUES OUTLINED IN THIS PROCEDURE OR USE THE EVA MANUAL.** • Drive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **C1** . Return vehicle to customer.
B11	TRANSFER CASE OUTPUT FLANGE RUNOUT		
	• Measure radial runout at transfer case front output flange. Runout should not exceed 0.203mm (0.008″). • Measure radial runout at transfer case rear output flange using dial indicator and companion flange runout gauge 205-319 and clamp plate 205-320 (Rotunda Tools). Runout should not exceed 0.203mm (0.008″). If either runout exceeds specification, call Borg-Warner at 1-765-286-6569 to receive a certified transfer case. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B12** . Return vehicle to customer.

TB-6152-A

Figure 5 - Article 00-14-3

Test Step		Result ▶	Action to Take
B12	REMOVE REAR DRIVESHAFT		
	• Remove the rear driveshaft from the vehicle marking the alignment to the axle and transmission with a paint pencil. • Drive vehicle in 4X4 high so vehicle is driven by the front driveshaft only. • **Does concern still exist?** **NOTE: VEHICLE WILL BE FREE TO ROLL WHEN REAR DRIVESHAFT IS REMOVED. ALSO, PROLONGED DRIVING (OVER 16km/10 MILES) WITHOUT THE REAR DRIVESHAFT COULD DAMAGE THE TRANSFER CASE CLUTCH OR VISCOUS COUPLING.**	Yes ▶ No ▶	GO to **B13** . GO to **B7** .
B13	REMOVE FRONT DRIVESHAFT		
	• Reinstall rear driveshaft, aligning the marks you made in Step B12. • Remove front driveshaft and examine it for damage. If damage exists, replace it. • Drive vehicle without front driveshaft. • **Does concern still exist?** **NOTE: PROLONGED DRIVING (OVER 16km/10 MILES) WITHOUT THE FRONT DRIVESHAFT COULD DAMAGE THE AWD TRANSFER CASE VISCOUS COUPLING.**	Yes ▶ No ▶	GO to **B14** . REPLACE driveshaft with proper service part. Redrive to verify repair. Return vehicle to customer.
B14	REPLACE FRONT AXLE		
	• Replace front axle with proper service part. Call Dana at 1-219-481-3332 to receive a certified front axle. • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **B15** . Return vehicle to customer.
B15	REPLACE TRANSFER CASE		
	• Call Borg-Warner at 1-765-286-6569 to receive a certified transfer case. • Redrive vehicle. • **Does concern still exist?**	Yes ▶ No ▶	GO to **A1** . Return vehicle to customer.

TB-6153-A

Figure 6 - Article 00-14-3

	Test Step	Result ▶	Action to Take
C1	CHECK ENGINE OIL LEVEL		
	• Check engine oil level. Be sure it is not overfilled. An overfilled engine sets off a vibration at 1000 and 2000 rpm. • **Is oil level correct?**	Yes ▶ No ▶	GO to **C2** . DRAIN oil and refill to correct level. Drive to verify repair. If concern is eliminated, return vehicle to customer. If concern still exists, GO to **C2** .
C2	CHECK FOR DTCs		
	• Bsing a New Generation Star (NGS) Tester, scan engine, transmission, Generic Electronic Module (GEM), and air suspension for possible Diagnostic Trouble Codes (DTCs). If DTCs exist, repair concern and retest vehicle. • **Is concern eliminated?**	Yes ▶ No ▶	Return vehicle to customer. GO to **C3** .
C3	ACCESSORY DRIVE TEST		
	• Examine engine for possible grounding of accessories. Inspect heater and A/C hoses, accelerator and speed control cables, and transmission cable for possible grounding. • Remove the accessories drive belt. • Run engine up to 3500 rpm. • **Does concern still exist?** **WARNING: DO NOT DRIVE VEHICLE WITHOUT ACCESSORIES DRIVE BELT.**	Yes ▶ No ▶	GO to **C4** . CHECK each component including the fan. Replace faulty component and return vehicle to customer.
C4	NEUTRALIZE ENGINE, TRANSMISSION, AND EXHAUST		
	• Perform neutralization of engine/ transmission/exhaust method in TSB 99-11 or Section 100-04 of the 2000 Workshop Manual. • **After performing neutralization procedure, is vibration gone?**	Yes ▶ No ▶	Return vehicle to customer. GO to **C5** .
C5	VIBRATION AT IDLE		
	• **Does condition exist at idle?** If vehicle has a 4.0L SOHC engine, install exhaust, Noise/Vibration/Harshness (NVH) brace, and clamps. • **Does concern still exist?**	Yes ▶ No ▶	GO to **A1** . Return vehicle to customer.

TB-6154-A

Figure 7 - Article 00-14-3